

***PRIMER EJEMPLO***

**>** *edo* :=  $y'' + \omega^2 y = 0$ ;

$$edo := \frac{d^2}{dx^2} y(x) + \omega^2 y(x) = 0 \quad (1)$$

**>** *cond* :=  $y(0) = u, D(y)(0) = 0$ ;

$$cond := y(0) = u, D(y)(0) = 0 \quad (2)$$

**>** *sol* := *dsolve*( {*edo*, *cond*}, *y*(*x*) );

$$sol := y(x) = u \cos(\omega x) \quad (3)$$

**>** *des* := *rhs*(*sol*);

$$des := u \cos(\omega x) \quad (4)$$

**>** *vel* := *diff*(*des*, *x*);

$$vel := -u \sin(\omega x) \omega \quad (5)$$

**>** *acel* := *diff*(*des*, *x*\$2);

$$acel := -u \cos(\omega x) \omega^2 \quad (6)$$

**>** *restart*;

**>**